9

Acquisition of the mental lexicon

In Chapter 7 of this textbook, we focused on the functional lexicon (the functional morphology) of L2 learners, arguing that it is of central importance to linguistic competence and performance because it contains the features that regulate grammatical meanings and syntactic transformations. In Chapter 8, we went on to describe and explain how learners convert these grammatical features into targetlike syntactic structures and meanings. In this chapter, our attention will be on words more generally speaking. Words are the basic building blocks of sentences that need to be represented and accessed in the mental lexicon when a sentence is comprehended or produced.[1] Word meanings are comprised of their own, idiosyncratic, lexical

[1] In discussing the mental lexicon, we talk about bilinguals. As a reminder, in the literature on the mental lexicon, "the term *bilingual* is used to refer to an individual with some competence in at least two languages without regard for the level of second language proficiency, regularity of second language use, or age or context of second

content (*door* is different from *gate*), but also of category features that organize them into classes of words with specific morphosyntactic behavior (both *door* and *gate* are concrete countable nouns, etc.). Words can be composed of several morphemes, some lexical and some functional, and lexical access may depend on that composition. Finally, some words may have literal as well as figurative meanings, which change the meaning of the whole sentence they participate in. In this chapter, I will attempt to provide a comprehensive outline of what we know when we know words and what aspects of that knowledge are easy or hard for L2 speakers.

9.1 Mental representation and access of lexical items

The study of the mental lexicon is one of the most vigorously developing areas of bilingualism. Several important discoveries have been made in the last 30 years that were unexpected from the point of view of our previous assumptions about the way words are organized and accessed in the bilingual and multilingual lexicon. We will see some earlier proposals for that organization, we will then review more recent behavioral data that seem to contradict the earlier models and to support one updated model. Following Dijkstra and van Heuven (2012), we will first discuss the facts about the bilingual lexicon on which most theoretical models agree, and which find support from recent neuroimaging (functional Magnetic Resonance Imaging, fMRI) and electrophysiological (event-related brain potential, ERP) studies. Next, we will discuss some issues on which researchers do not agree yet.

But before we go on to discuss the findings, let me introduce the tasks that are used in studying the mental lexicon. In a *lexical decision task*, participants are shown lexical items in isolation (usually on a computer screen) and asked to decide whether these are legitimate words in a language. In a *semantic task*, they are asked whether words are related in meaning. In a *naming task*, they are shown pictures and asked to come up with a name of an object as quickly and accurately as possible. Reaction times, ERP effects, and blood flow can be measured while research participants are performing

language acquisition" (Tokowicz 2014: 4). Very often in this literature, the term *bilingual* is used to refer to multilingual individuals as well, in the way we described the term in Chapter 5.

these tasks. In a *primed lexical decision*, words can be preceded for a very short time (say, 250 milliseconds) by a certain other word, or part of a word, before the actual lexical decision task is performed. This previous exposure is referred to as "priming." It has been shown that participants are faster to respond to words when they are primed with a semantically related word. For example, participants are faster to confirm that *nurse* is a word of English when primed with *doctor* than when primed with *butter*. In studies using the *visual world paradigm* in eye tracking, an array of objects is presented either physically or on a computer screen, and the individual's eye movements are tracked. These experimental tasks and the measurements they yield allow researchers to ascertain the relations of words and morphemes in the mental lexicon.

9.1.1 Bilingual lexicon representation models

Generally speaking, memory representation models try to describe the way that word forms and meanings are represented in memory, and how the words in the two languages and their meanings are related to each other. One class of models considers two levels of representation within the bilingual memory: that is why they are called "hierarchical." For example, the Revised Hierarchical Model (Kroll and Stewart 1984) postulates that the words of the two languages reside in separate memory systems, but that concepts are stored in a common memory system. The model also proposes that the interconnections between these representation systems, or stores, are asymmetric: L1 words have a stronger connection to the common meaning store than L2 words. Because of this asymmetry between the systems, the so-called *translation equivalents* (words with roughly the same meaning) are crucial in the beginning stages of acquisition. They mediate between the L1 and the L2 at the lexical level, until learners are skilled enough to access the L2 meanings directly.

Another class of models is called "interactive." Prominent among them is the Bilingual Interactive Activation Plus model (BIA+) (Dijkstra and van Heuven 2002), which postulates organization of the mental lexicon on three levels: one level of conceptual representation (the lemma level),[2] a lower

[2] Psycholinguists distinguish two separate aspects of the mental lexicon entries, named *lexeme* and *lemma*. The lexeme stores morphological and formal information about a word, such as the different versions of spelling and pronunciation of the word; the lemma

level containing the words, and the lowest level containing orthographic, grammatical, and phonological word features. In accessing a word, orthographic information (for example, if a word is read) activates the orthographic word representation, and activation then spreads to the higher levels of representation.[3] In an earlier instantiation of the model (BIA, van Heuven, Dijkstra, and Grainger 1998), there was a fourth level included, the language node. Since words are connected to the language node to which they belong, the model contains a way for making a language decision (which language am I speaking now?). Use of one word over another can raise the activation of one language while inhibiting the other. Using this mechanism, the model can account for the effects of relative second language proficiency. Later findings made the proponents of the BIA+ model adjust the language node as a task/decision component affected by extra-linguistic factors.

Both the hierarchical and the interactive models assume that the lexicon entries of the two languages are not located in separate parts of the brain but are instead functionally interconnected and have a common conceptual-semantic basis, as discussed below.

9.1.2 The bilingual lexicon is integrated across languages

After several decades of experimental research, it is widely accepted that the lexical items of the different languages in a multilingual individual's brain are integrated and stored together. It is not the case that the lexical items for the L1 and the L2 reside in different areas of the brain, or even in separate stores. This is true for orthographic, phonological, and semantic representations. The implication **from an integrated mental lexicon** is that "lexical representations belonging to different languages affect each other in the same way as representations within a language" (Dijkstra and van Heuven 2012: 453). To use a spatial metaphor, representations are stored closer together the more similar they are, irrespective of the language they belong to. This is true for both phonological and orthographic representations.

stores semantic and syntactic information about a word, such as word category and the meaning of the word. Research has shown that the lemma develops first when a word is acquired into a child's vocabulary, the lexeme develops later with repeated exposure.

[3] The model is based on the interactive activation model of McLelland and Rumelhart (1981).

This brain organization has been supported with consistent findings from brain imaging studies. When the degree of L2 proficiency among speakers has been high, and comparable to the L1, the same brain areas have been activated in L1 and L2 processing.[4] The activations found for the L2 were similar, if not identical, with those utilized by the same individuals in L1 lexical item retrieval. Moreover, this happened irrespective of the degree of difference between languages in orthography, phonology, or syntax (Chee, Tan, and Thiel 1999). These findings are also consistent across various tasks: lexical decisions and semantic judgments. As we saw before, at lower proficiency levels, L2 users were utilizing wider areas in the cortex to compensate for the less efficient access. Factors affecting the location of activations, apart from proficiency, are age of acquisition of the second language and (type of) language exposure.

9.1.3 The bilingual lexicon is accessed in a language-independent way

The second important discovery coming from recent research on the mental lexicon has to do with the lexical search and access. To cite the engaging example from Tokowicz (2014: 1), imagine you speak Spanish and English, and your friend asks you in English to hand her one of a few items within your reach. The beginning sound of the word is /k/ and that sound is compatible with a cup and a spoon (*cuchara* in Spanish). What happens next? The question is whether, when we are confronted with a word, we first access the lexicon of one language and then the other, or there is a **parallel search through both/all languages**, words not being organized primarily on the basis of language, but in semantic and phonological fields on the basis of frequency.

The evidence comes on the side of the second option: alternatives in each language are activated when only one language is being read/heard or produced. For example, using the visual world paradigm, Spivey and Marian (1999) and Marian and Spivey (2003) have shown that bilingual listeners look more to the competitor item (the spoon in the example above) than to control items that start with other sounds in Spanish and English. In another example, presenting English words such as *blue* to a Dutch-English bilingual will activate not only other English words such as *clue* and *glue* but

[4] These are the left frontal and temporal-parietal areas. For a review, see Abutalebi and Della Rosa (2012).

also Dutch words like *blut* and *blus* (Dijkstra and van Heuven 2002: 454). Such findings suggest that all the available lexicons are briefly activated in the beginning of the lexical search. This view has been dubbed the "language-nonspecific access." The so-called neighborhood effect provides another set of evidence against selective access (van Heuven et al. 1998, Jared and Kroll 2001, Dijkstra 2003): if a word has many "neighbors," words that differ in one letter or sound as in the example above, it will take longer to recognize that word in a lexical decision task. Of course, if a bilingual could access words selectively in one of the two languages, such interference would not surface. It has been determined that neighbors from the second language do increase the reaction times for accessing L1 words, which shouldn't happen if the language stores were separate. These findings have been interpreted to support non-selective access, since lexical items of both languages obviously compete in the activation process.

Neuroimaging studies of interlingual homographs also offer support for language-nonselective access. Interlingual homographs are words that have the same orthography but are separate lexical items with different meaning in the L1 and the L2. One such word is *ramp*: the word has a few meanings in English but in Dutch it means "disaster." Behavioral studies often provide evidence for interference when processing such words. Van Heuven, Schriefers, Dijkstra, and Hagoort (2008) investigated the source of interlingual homograph conflicts with two fMRI experiments. In one experiment, brain responses of Dutch-English bilinguals were examined when performing an English lexical decision task (is this a word of English?). In the other experiment, they were given a generalized Dutch-English task (is this a word?) In both cases, co-activation was expected. However, a conflict should arise only in the English lexical decision task, because a Dutch word in that experiment should be rejected. As expected, imaging showed a response conflict only in the English lexical decision task. In both tasks, co-activation was also established. Evidence from studies using cognates, homophones, and cross-linguistic priming follow a similar logic as in the interlingual homograph studies, and the evidence for non-selective activation of lexical items is overwhelming.

9.1.4 Language exposure and use affects the activation of words in the lexicon

This assertion is hardly surprising. When words are used more often in daily life, hence are more frequent, they become established more strongly in long-term memory and are easier to access when we need them. This is true of words from the native language as well as any subsequent language. Based on that premise, it is logical that relative L2 proficiency, age of acquisition, and language use are all factors that affect accuracy of recognition and speed of access. However, consider that frequency is to some extent relative and individual-based. Even frequent words may be less frequent in the usage pattern of a non-proficient bilingual, and it is this personal frequency that matters in lexical access. Cognitive neuroscientific studies confirm that proficiency affects the activation pattern of the L2 lexical access (relative to the L1 access). We will cite here only one of many studies, Chee, Hon, Lee, and Soon (2001), which established that in making semantic judgments in their L2 English, Chinese participants showed a larger activation area (in the left prefrontal and parietal locations, as well as additional activation in the right inferior frontal gyrus). A general finding of such studies is that more extensive or greater levels of brain activation can be detected in the less proficient language of bilinguals. This is true not only of word processing but more generally of language processing. The more the language is practiced, the more efficient processing becomes.

9.1.5 Language context may not affect bilingual language activation

How do speakers choose the words that they are going to use in a sentence while producing speech? In other words, how does language selection occur? Kroll, Gullifer, and Rossi (2013) review cues for language selection, the most interesting and obvious being the surrounding linguistic context (which language is being spoken). If bilinguals are not very good at language selection when accessing words, we would expect to see pervasive errors in bilinguals' speech, in the form of using lexical items from the other language often and haphazardly. This does not happen on a regular basis, so bilinguals must have a mechanism for overcoming that influence.

However, there is little doubt that this influence exists. For example, Van Asche et al. (2009) reported cognate effects when reading sentences in the native Dutch of English-Dutch bilinguals. Although the sentences appeared only in their native language, there was a persistent effect of English, their L2, on their L1. Such results suggest that even when processing one language, the influence of the other language, or languages, that the bilinguals know remains very much in evidence, and in action. This discovery is particularly surprising when the second language knowledge influences the functioning of the native language. However, that discovery logically leads us to postulate that bilinguals must constantly inhibit one language when speaking the other. Another conclusion is that a great deal of cognitive control is necessary for the lexical performance of a bilingual to be accurate. This last claim leads us to the next section.

9.2 Inhibition of one language to speak another

The third discovery, which is true of language processing in general but was first formulated about and exemplified in the mental lexicon, is that **it is virtually impossible to switch off the language not in use.** The parallel activation of a bilingual's two languages can be observed in reading, listening, and in planning speech (Costa 2005, Dijkstra 2005, Kroll et al. 2006, Marian and Spivey 2003, Schwartz, Kroll, and Diaz, 2007). Thus the main challenge for bilinguals trying to access the vocabulary of one language is actually suppressing the vocabulary of the other language(s) she knows.

Let us look at this inhibition process in more detail. David Green's Inhibitory Control model (Green 1998) was designed to explain how bilinguals regulate and exercise control over which language they speak in. Within the area of the mental lexicon, Green and other researchers (Guo Liu, Misra, and Kroll 2011, Misra, Guo, Bobb, and Kroll 2012, among others) have proposed that there is competition for lexical selection, with candidates from both languages available, and subsequent inhibition of the language not in use. When this language happens to be the stronger or more dominant language, the processing costs of inhibition are greater. Interesting support for this claim comes from language switch costs.

Meuter and Allport (1999) used a task in which bilinguals had to read aloud numerals, and then switch to another language unpredictably, cued by a background color. The researchers established that bilinguals suffer larger switch costs when switching from the L2 to the L1 than in the opposite direction. Although this may seem counterintuitive at first, it is explained by a spillover effect. In order to name objects in the L2, the more available and activated L1 is should be suppressed, and that inhibition spills over into the subsequent naming trial, although the word to be named is in the L1.

Current experimental methods such as ERPs allow us to investigate the course of planning during actual or tacit naming. Compared to the brain imaging methods that monitor blood flow, ERPs are capable of tracking very fast processes, on the order of milliseconds. In this respect, they permit more time-sensitive analyses of ongoing cognitive processes. One such ERP component is the N2, (sometimes also referred to as the N200), which reaches its peak at approximately 300 milliseconds after stimulus onset. Although researchers are not yet in agreement on what the N2 exactly signifies (inhibition of a response or a response conflict), there is relative consensus on its being an index of a control process. According to the Inhibitory Control model (Green 1998, Abutalebi 2008), bilinguals select the words of the language in use by inhibiting the activation of the other language. ERP studies using the switching experimental paradigm found a significant N2 effect for switch trials, hence supported the Inhibitory Control model. Gollan and Ferreira (2009) also supported the model with a voluntary switch paradigm in a naming task, comparing switch trials to single-language trials. The cost to naming times for the voluntary switching condition was greater for the dominant language. Gollan and Ferreira (2009) concluded that inhibition of the first language is a standard part of production in the second language, even for proficient bilinguals.

The competition that is hypothesized to be a result of selecting lexical candidates for production is regulated by a network of brain structures associated with executive function and inhibitory control (e.g., the left prefrontal cortex and the anterior cingulate cortex, ACC). It is precisely this constant inhibition (exercised through executive control) that Bialystok credits with giving bilinguals a cognitive advantage over monolinguals. This constant parallel activity produces changes in the bilingual brain. A recent study, Abutalebi et al (2012) showed that the ACC appears to be changed by bilingualism. When placed in a context where they are expected to monitor conflict, not only do bilinguals outperform monolinguals, but they also appear to need fewer cognitive resources to do it. Bilinguals have also been shown to outperform monolinguals on task switching and ignoring irrelevant information. It could be the case that components of the brain that enable constant monitoring and control of the bilingual functioning spill over more general cognitive functioning. If this is indeed the case, as evidence suggests, there is a big fat cognitive bonus of bilingualism.

Teaching relevance

In this section, we have been discussing lexical access and usage. The results of the studies we reviewed cannot be directly used in language classrooms. However, there is an interesting recent finding in this area that is a dramatic demonstration of the impact teaching a second language has on the brain of even very beginning learners. The seminal study of McLaughlin, Osterhout, and Kim (2004) provides evidence that the brain starts processing and learning at the very first exposure to the second language, and crucially, before any learning is detected in behavior changes. ERPs were recorded at 14, 53, and 138 mean hours of instruction of French to native English speakers. The control group was monolingual. The participants performed a primed lexical decision task. The prime target pairs were either semantically related words (*chien–chat* 'dog–cat'), semantically unrelated words (*chien–table* 'dog–table') or word–nonword pairs (*chien–nasier* 'dog–nasier'). Even after the earliest exposure, 14 hours of French lessons, learners showed a significant modulation of the N400 ERP component, which references lexical semantic processes, although their behavior was the same as that of the non-learners. Thus, this is a positive indication that learners' minds are indeed changing, even if it doesn't appear to be the case in behavior. Even at 14 hours' exposure, learners have already started to build a mental grammar!

In summary, the processes of lexical access seem to be largely similar in first and second language acquisition and use. The areas of the brain utilized in processing also seem to be largely overlapping, and those are the main language-processing areas. Proficiency seems to drive the crucial L1–L2 differences, in the sense that greater engagement of brain areas appear to be related with processing language that is not mastered to a nativelike level. That activity may reflect executive control over access to short-term or long-term memory representations such as grammatical, phonological, or lexical representations. The main idea is that a low-proficient second language will be processed through neural pathways related to controlled processing, that is, with the active engagement of the brain areas related to cognitive control. Such control is necessary to inhibit the stronger or native language when we are speaking a weaker language or a second language. On the other hand, a proficient second language is processed in a more nativelike manner, that is, with a higher degree of automaticity and efficiency.

9.3 Morphological decomposition in the lexicon

We turn to a different topic now, namely, what the processes are that go on within the L2 lexicon itself when we store words in the mental lexicon. At issue is whether words are stored whole, or decomposed into morphemes in storage and assembled online when needed. True to the main focus of this book, we will be mostly interested in the processing of functional morphology. Michael Ullman's Declarative/Procedural model is the one relevant here.[5] The model argues that processing a language, even one's native language, involves two different brain systems: the *declarative and procedural memory systems*. The former is a lexical store of memorized words that depends on our declarative memory, the part of the brain that we use to recall and recite memorized information such as the state and country capitals. The latter is a combinatorial mental grammar that uses productive rules, working with memorized units such as words and morphemes to produce novel but legitimate combinations of these.

The two memory systems are not only functionally separate but also rooted in different brain structures. Given these assumptions, Ullman (2001, 2004) argues that maturational changes occurring during childhood/adolescence lead to the weakening of the procedural and a boost of the declarative system. Thus L2 learning and processing in adulthood are much more dependent on the lexical memory system and not so much on the procedural system, compared to L1 processing. Of course, these claims are not to be taken as categorical. If adult L2 learners could *only* use their declarative memory, they would not be able to produce a single sentence that they had not heard before. (Almost) everything is a matter of degree.

What would Ullman's model predict with respect to how adult L2 learners process functional morphology? Keep in mind that if words are stored in the mental lexicon decomposed into affix and stem, then they have to be composed on the fly, as needed in production or comprehension. Thus morphological decomposition engages the procedural memory system while whole-word lexical storage can be serviced better by the declarative memory system.

[5] Michel Paradis's neurolinguistic account of SLA is largely in agreement with Ullman's model (Paradis 2004, 2009).

Whether bilinguals decompose regularly inflected words or not can be studied with a lexical decision and lexical priming tasks. In morphological (not semantic) priming, the prime is morphologically related to the target word and it flashes on the screen for a very short time (e.g., 60ms). If the past participle *work-ed* is primed with the stem *work* and access is facilitated (that is, *faster* as compared to a foil such as *play*), then this is considered evidence for morphological decomposition in the mental lexicon: a part of the word primes the whole. The most popular version of this task uses "masked" priming, where the prime is masked by symbols such as ###### before and after it appears, to diminish the prime's visibility and make its effect over the lexical decision even more automatic and unconscious.

In a series of experiments, Neubauer and Clahsen (2009) investigated lexical access to German participles ending in *–t* (regular) and *–n* (irregular) by German native speakers and very highly proficient Polish learners of German. The lexical decision task uncovered an effect of frequency in both groups: more frequent grammatical forms were recognized more quickly. However, this frequency effect was only attested for the irregular participles but not for the regular participles in the natives. In contrast, learners showed frequency effects for both regular and irregular participles. This outcome suggests that L2 processing relies more on memory storage than L1 processing, supporting Ullman's model. Natives store regular participles in decomposed form (stem and affix), so frequency is not so important for fast lexical access. In contrast, learners presumably store participles whole, without decomposing them into stem and affix. In accessing them, they use declarative memory, and so word frequency is the most important factor for the speed of that access. In the priming tasks of the same study, Neubauer and Clahsen found that the full stem primed the regular participles in natives, but not in learners, in support of the lexical decision findings. Similar findings come from another experiment by Silva and Clahsen (2008), this time from the access of derivationally composed words. In the priming tasks, *bitter* did not prime *BITTER-NESS* in the learners, indicating that they store each word whole and do not decompose.

However, morphological decomposition in bilinguals has been experimentally attested as well. The first study to document decomposition is Diependaele, Duñabeitia, Morris, and Keuleers (2011). It explored morphological processing in a native and nonnative language using derivational relationships (e.g., *walker-WALK*) in a masked priming lexical decision task with semantically transparent and opaque derivational relationships, as well as form-related items. For example, in *walker-WALK*, the derivational relation is transparent, in *corner-CORN* the derivational relationship is opaque; while *freeze-FREE* are only related in form. A group of native English speakers and two groups of Spanish-English bilinguals of varying levels of proficiency participated. Interestingly, results showed similar priming patterns for the native participants and the two groups of bilinguals (i.e., no significant differences in the magnitude of the morphological priming effects). Derived words from a nonnative language were decomposed early and accessed through the constituent morphemes in a fashion similar to that from a native language. In addition, this experimental study suggested that morphological decomposition of polymorphemic words may be an automatic process that does not depend on the proficiency of the reader in the language at stake.

Similar conclusions were reached by other recent studies.[6] Gor and Jackson (2013) set out to challenge the view that L2 learners do not decompose morphology in their mental lexicon. The also used masked priming tasks; their participants were learners of Russian as an L2 at three levels of proficiency. Russian is a morphologically rich language, so it is common for words to contain a root, derivational and inflectional morphemes, in a Russian doll fashion. The example in (1) illustrates the root and the derivational suffix that make up the stem, to which the inflectional morpheme is added.

(1)	rabot	– aj	– u	'I work'
	root	suffix	inflection	

Using regular, semi-regular, and irregular verb forms, the researchers found that morphological decomposition is pervasive and automatic at all three proficiency levels, at least for the outside inflection morphemes. Secondly, bilinguals access the stem representation either directly or by further decomposing the stem into root and suffix. This second stage of lexical processing is gradually acquired by late learners, from productive and less complex stem allomorphs to unproductive and more complex stem allomorphs.

[6] De Diego Balaguer et al. (2005), Duñabeitia, Dimitropoulou, Morris and Diependaele (2013), Gor and Jackson (2013), Feldman et al. (2010). See the special issue on recent advances in morphological processing, *Language and Cognitive Processes*, 28, 7, 2013.

Together with the results of the studies mentioned above, Gor and Jackson's findings suggest that L2 speakers do decompose polymorphemic words. Frequency, regularity, morphological richness of the native language, and L2 proficiency are all important factors in this decomposition process. At the same time, there is no doubt that the arc of development is as described by the declarative/procedural model. Learners start from storing whole words, but as proficiency increases, they are capable of using their procedural system more and more. Gor and Jackson (2013:1066) make a very valid point: for some highly inflected languages such as Russian, using only declarative memory to store enough low-frequency inflected words undecomposed may be impractical. Almost immediately after the beginning stage, when L2 learning relies on memorizing whole words and chunks, learners have to start decomposing many inflected words for which they might not yet have whole-word stored representations. The bottom line of this discussion is that, as with almost all linguistic processes, morphological decomposition in the mental lexicon is available to L2 speakers, but it could also depend on the language being acquired.

9.4 Argument structure

In this section, I turn to explorations of lexical semantics acquisition, both from the point of view of generative grammar and of cognitive semantics. The focus will be on cases where there is a lexical semantics mismatch between the native and target language in argument structure. To remind the reader, the array of arguments that is necessary for the expression of a verb meaning is considered its argument structure. Generative studies of lexical semantics explore the relationship between argument structure, lexical meaning, and overt syntax. The theoretical foundation for much of this work is the semantic decomposition of lexical items into primitives or conceptual categories such as Thing, Event, State, Path, Place, Property, Manner, etc. (Jackendoff 1990, Pinker 1989, Grimshaw 1990). For example, the meaning of the verbs *kill* and *break* can be represented as made up of the following semantic primitive concepts (in caps) that are seen as making part of the meaning of other verbs as well:

(2) kill = CAUSE [BECOME [NOT [alive]]
    break = CAUSE [BECOME [NOT [whole]]

Other theoretical proposals establish concrete linking rules between syntactic positions and thematic roles (Levin and Rappaport Hovav 1995, 2005). Verbs can appear in different configurations with their arguments, and when they do, we call this a syntactic alternation. The sentences in (3) exemplify the so-called *causative–inchoative alternation*.

(3) a. John broke the vase. (causative)
b. The vase broke. (inchoative)

Is the meaning of the verb *break* the same in the two sentences, or does it change slightly? In (3a) the verb appears with an Agent as subject and a Theme as object. However, in (3b) the Theme is linked to subject position. To take another example from the English dative alternation, it may seem that the two syntactic frames that alternate, the double object and the dative, are completely semantically equivalent, but it is not really so.

(4) a. Mary sent John the package. (Double object syntactic frame)
b. *Mary sent Chicago the package.
c. Mary sent the package to John. (Dative syntactic frame)
d. Mary sent the package to Chicago.

In the double-object frame exemplified in (4a) and (4b), it is necessary that at the end of the event of sending, the Goal is actually in possession of the Theme, the package. While this is possible in the case of a Goal such as John, it is not possible for a Goal such as the city of Chicago. A city cannot possess a package.

Note that the dative syntactic frame exemplified in (4c) and (4d) might actually confuse the learner of this alternation. Why is Chicago an acceptable Goal in (4d) but not in (4b)? This is because, as linguists have argued, the dative alternation does not have the requirement (constraint) that the Goal should be in possession of the Theme: the latter could have just been sent in the direction of Chicago. In other words, the difference between the two constructions is a matter of the entailment of possession in the double object construction, and lack of such entailment, in the dative construction. The clincher in this line of argument comes from a slight change of meaning. If you imagine that two office workers are talking about a package that has to be sent to the head office of their company, which happens to be in

Chicago, then sentence (4b) becomes acceptable. In that new construal, Chicago stands for the head office, not the city, and presumably there are people in the main office who can be in possession of the package.

> Before going forward, discuss the following examples. What is going on here?
>
> (i) The noise gave Terry a headache.
> (ii) * The noise gave a headache to Terry.

Furthermore, it is not the case that every verb that can take a dative argument alternates with a double object construction. For example, the sentence in (5) is unacceptable.

(5) *Sam pushed Molly the package.

Why is the double object frame available for *send* but not for *push*? Why is the possession entailment available only for the double object construction? To address learnability challenges such as these, Juffs (1996) and Montrul (2001a) have argued that in the mapping from the lexicon to syntax, there is a logical problem for L2 acquisition: L2 learners have to discover the possible mappings between meaning and form in the absence of abundant evidence. Both the patterns exemplified in (4) and in (5) constitute a Poverty of the Stimulus learning situation: superficial analogy between two constructions leads to wrong assumptions about acceptability. There is rarely a one-to-one relationship between syntactic frames and available meanings. Though some mappings may be universal, there is a lot of cross-linguistic variation in this respect. It is very easy to overgeneralize lexical alternations that appear in the native language (thus assume that (5) is acceptable), and it is not obvious how learners can retract from such overgeneralizations. Such research questions inspired a lot of experimental studies in the 1980s and through the early 2000s. Mazurkewich (1984), White (1987), Bley-Vroman and Yoshinaga (1992), and Whong-Barr and Schwartz (2002) studied the dative–double object alternations. Findings of these studies are largely consistent with the claim that L2 learners initially adopt L1 argument structures.

Cross-linguistic differences in conflation patterns (i.e., what primitives of meaning are conflated in a verb) as illustrated by the Chinese and English examples in (6) and (7) were taken up by Juffs (1996) and Inagaki (2001). In English, verbs that express psychological states can appear with a Theme subject and an Experiencer object, as in (6) while in Chinese this usage is not possible, as (7) illustrates.

(6) The book disappointed Mary.

(7) *Nei ben shu shiwang le Zhang San
that CL book disappoint PERF Zhang San
'That book disappointed Zhang San.'

Of course, Chinese can express a similar meaning, but with another, periphrastic, construction (Juffs 1996). However, this gap in argument structures presents difficulties for L2 acquisition: in learning English, Chinese speakers have to learn the availability of constructions such as (6); in learning Chinese, English speakers have to unlearn, preempt, or acquire the fact that (7) is unavailable. Juffs tested Chinese native speakers learning English in China on acceptance and production of psychological, causative, and locative verbs that do not have an equivalent in Chinese. He related these three types of verbs to a lexical conflation parameter: English verbs allow the semantic primitives CAUSE and STATE to be conflated in the same verbal root, while Chinese does not. Results suggest that learners at low to advanced levels of proficiency are sensitive to the conflation pattern of English, having acquired structures unavailable in their native language.

Another study examining conflation differences, Inagaki (2001), is a bidirectional (English to Japanese and Japanese to English) study of motion verbs with Goal PPs. Using the same test in both learning directions, a picture followed by sentences to be judged for appropriateness, Inagaki found that there is evidence for directional differences in acquiring L2 conflation patterns. English learners of Japanese overgeneralize their native pattern, which is unavailable in Japanese, but Japanese learners of English have no trouble learning the new pattern on the basis of positive evidence. While supporting L1 transfer in learning lexical form–meaning mappings, such results also highlight the issue of the availability of negative evidence in L2 acquisition. See Exercise 9.2 for more details of this study.

Unaccusativity presents another subtle semantic difference between classes of verbs that relies on argument structure mapping. Intransitive verbs are verbs that allow only one argument. However, not all intransitive verbs are created equal. Linguistic tests distinguish two classes of intransitive verbs, with different syntactic behavior and subtle differences in meaning. Unaccusative verbs are intransitive verbs whose only argument is a Theme, or underlying object (e.g., *fall, arrive*), while unergative verbs are intransitives with an Agent argument (e.g., *dance*, *laugh, sneeze*).[7]

(8) a. John danced.
    b. John danced a jig/a polka/a little dance.

(9) a. John fell.
    b. *John fell a fall/a nasty fall.

The subject of the unergative verb *dance* is an Agent, that is, in control of the event and causing it. Although the verb is intransitive in (8a), it can also appear with a cognate object, or with names of certain dances, as (8b) illustrates. On the other hand, the only argument of the verb *fall* is a Theme: John undergoes an action that he did not cause. Falling happens *to* you precisely because you are not in control. This analysis is supported by the impossibility of adding more Theme arguments as objects in (9b). The unaccusative–unergative distinction is a universal distinction in the sense that all languages have the two classes of verbs. Moreover, this distinction can be related to verb meaning: unaccusatives express processes that can happen autonomously, spontaneously, without external influence, while unergative verbs express changes which have some (possibly unexpressed) external causer.

Unaccusativity can be syntactically manifested in different ways in different languages. For example, some languages such as German distinguish between these verbs through auxiliary selection: unergatives take the equivalent of *have* and unaccusatives take the equivalent of *be,* when they are used in complex tenses such as the perfect or the *passé composé*. In addition, in French only the participles of unaccusative verbs agree with the subject in gender (13–14).

(10) Sie **hat** gearbeitet/telefoniert/gekocht/geraucht. [German]
     she has worked/telephoned/cooked/smoked

[7] These names are terms that come from the linguistic literature and may not be meaningful to the reader. They should be remembered as terms.

(11) Sie **ist** gestorben/alt geworden/runtergefallen/aufgewacht.
she 'is' died/become old/fallen down/woken up

(12) Elle **est** mort-**e** hier. [French]
she is died-FEM yesterday
'She died yesterday.'

(13) Elle **a** écrit pendant toute la journée.
she has written during the whole day
'She wrote the whole day.'

Hirakawa (1999) tested knowledge of the unaccusativity verb distinction in the interlanguage of Chinese and English native speakers learning Japanese. One of the properties she investigated was whether learners were aware of the fact that combined with different classes of verbs, the adverb *takusan* 'a lot' refers to different arguments. The number sign (#) signals that the interpretation expressed in the sentence is not available.

(14) Takusan yon-da. [Japanese]
a lot read-PAST
'Somebody read a lot of things.'
#'A lot of people read something.'

(15) Takusan ason-da.
a lot play-PAST
'Somebody played a lot.'

(16) Takusan tui-ta.
a lot arrive-PAST
'A lot of people arrived.'

Hirakawa took advantage of a particular property of Japanese: that the overt subject and object can be dropped (if recoverable from context). The reading of (14), a sentence with a transitive verb but no overt argument, is that somebody read a lot of things but not that a lot of people read one thing. This is how we know that *takusan* modifies the underlying object, or Theme argument. The reading of (15) with an unergative verb is along the same lines. However, the meaning of (16) with an unaccusative verb is that a lot of people arrived, not that someone did a lot of arriving. Thus *takusan* modification distinguishes unaccusative and unergative verbs in Japanese. However, recall that the verb distinction itself is universal, only its markers vary across languages (see examples (8)–(16) above). Hirakawa used a Truth Value Judgment Task with pictures to probe underlying linguistic knowledge. Again as in Juffs' and Inagaki's studies, the findings indicated successful acquisition. Those learners who had acquired the meaning of (14) were also accurate on the distinction between (15) and (16). Hirakawa argued that her participants displayed knowledge of deep unaccusativity.

Finally, in a series of studies, Montrul (2000, 2001b) studied acquisition of the causative–inchoative alternation and its relation to inflectional morphology, in L2 Spanish, L2 Turkish, and L2 English. We exemplified the causative–inchoative alternation for English in (3): the verb forms are the same, no matter whether they are in a causative or an inchoative frame. As can be seen from the examples below, in Spanish (17) the inchoative member of the alternation is morphologically marked with a reflexive particle *se*. On the other hand, in Turkish, both members of the alternation can be marked, but with different morphology: causative in (18) and passive in (19). The examples are from Montrul (2000).

(17) a. El enemigo hundió el barco.
the enemy sank the ship
'The enemy sank the ship.'

b. El barco **se** hundió.
the ship REFL sank
'The ship sank.'

(18) a. Düşman gemi-yi bat-**ır**-mış.
enemy ship-ACC sink-CAUS-PAST
'The enemy sank the ship."

b. Gemi bat-mış.
ship sink-PAST
'The ship sank.'

(19) a. Hırsız pencere-yi kır-dı.
thief window-ACC break-PAST
'The thief broke the window.'

b. Pencere kır-**ıl**-dı.
window break-PASS-PAST
'The window broke.'

The gist of Montrul's findings is that acquisition of argument structure alternations crucially depends on the argument-change-signaling morphology. Learners who speak a language where alternations are overtly marked in the morphology (a suffix signaling the causative in Turkish, a clitic signaling the inchoative Spanish) are more sensitive to these alternations in a second language than learners whose native language has no overt morphological reflex of the alternation (English). Such outcomes suggest that overt morphology actually facilitates the acquisition of argument structure alternations, highlighting the logical problem of lexical meaning acquisition in some languages that are poor in such morphology. (And another confirmation that functional morphology is crucial in acquisition.)

Very similar learnability issues have also been studied within the framework of cognitive semantics, that is why we will discuss some of those results, too. Within cognitive semantics, the influential work of Talmy (1991, 2000) has provided inspiration for much L2 acquisition research. Talmy (1991) suggests that languages can be divided into two typological groups depending on how Path of motion is lexicalized: in the verb (verb-framed) or outside the verb (satellite-framed).

(20) Tama-ga saka-o kudaru
ball-NOM hill-ACC descend
'The ball descends the slope'

(21) The ball rolls down the hill.

In (20), a prototypical example from Japanese, the event Path is lexicalized in the verb *kudaru* 'descend.' In (21), a corresponding prototypical example from English, Path is lexicalized in the so-called satellite, the verb particle *down*. The reader should be reminded of a very similar property tested by Inagaki (2001). It is important to note a methodological difference: while generative studies on lexical meaning typically employ experimental methods for assessing comprehension, cognitive semantics studies typically scrutinize language production, either elicited or from corpora. In a series of studies, Cadierno and colleagues[8] investigated this type of lexicalization pattern in L2 English and L2 Spanish by learners speaking typologically similar and typologically different languages. Findings indicate that even though intermediate and advanced L2 learners are generally able to develop the appropriate L2 lexicalization patterns, they still seem to exhibit some L1 transfer effects. In particular, in Cadierno and Robinson (2009), the two groups of learners (Danish-to-English and Japanese-to-English) demonstrated possible successful acquisition, L1 effects, as well as some effects of task complexity. Brown and Gullberg (2010) argue that learning a second language affects the lexicalization pattern employed in the native language, so language transfer is not only unidirectional.

In a corpus-based study, Lemmens and Perez (2010) investigate the use of Dutch posture verbs (equivalents of *stand, lie*, and *sit*) by French learners of Dutch. The authors come to the conclusion that the interlanguage system should be treated as a linguistic system in its own right, and that it shows both errors due to L1 transfer, in this case underuse of posture verbs, as well as errors due to overextension of the pattern that learners have acquired in the target language. Looking at the usage of similar verbs (*put* versus *set/lay*) in speech analysis as well as in gesture, Gullberg (2009) employed an elicited production task of describing placing events that English native learners of Dutch have just seen on video. Their production was video-recorded. Along the lines of Lemmens and Perez, Gullberg also argues that her subjects show some sensitivity to the target semantic patterns of the L2, although they are not targetlike.

In sum, research within cognitive linguistics more often scrutinizes production rather than measures comprehension as generative lexical semantic research does. Nevertheless, both research traditions come to very similar conclusions: lexical semantics presents significant difficulties to L2 learners. This is particularly evident when learners have to restructure their lexical knowledge in such a way as to acquire new markers of verb alternations or new argument structure mappings. However, these difficulties are not insurmountable and successful acquisition is attested in the majority of studies.

[8] Cadierno (2004, 2008), Cadierno and Ruiz (2006), Cadierno and Robinson (2009).

9.5 Transfer of reference

At the end of this chapter, I would like to highlight another area of lexical acquisition that has garnered even less research attention than argument structure alternations, but nevertheless may present substantial difficulties to learners. It goes by many names: metonymy,[9] polysemy, figurative language, enriched composition,[10] and transfer of reference.[11] What is involved is a type of implicature, but on a lexical level: what is said is different from what is actually meant. Let's take a classic example from this literature (Jackendoff 1992, Nunberg 1979, 1995).

(22) The ham sandwich wants another coffee.
[*meaning:* the person contextually associated with a ham sandwich]

Now, this sentence is plainly nonsensical if uttered out of the blue and in just any situation. But if it happens in a diner, and if this is a conversation between two waiters, the sentence suddenly makes sense: they are referring to the person contextually associated with a ham sandwich, maybe because s/he ordered one, or is eating one. Let's consider two more examples:

(23) Ringo was hit in the fender by a truck when he was momentarily distracted by a motorcycle.
(Nunberg 1995, modified from Jackendoff 1992)
[*meaning*: the car Ringo was driving]

(24) [One flight attendant to another]:
Ask seat 19 whether he wants to swap.
(Markert and Nissim 2006: 159, British National Corpus)
[*meaning*: the person sitting in seat 19]

The literal meaning of the underlined phrase, determined by the meanings of the constituent words, is enriched to a pragmatically determined additional meaning given in square brackets under the examples.

Of course, metonymy is a well-established mental process, whereby naming some entity (activity, person, thing, time period, etc.) is interpreted to stand for a related entity. Typical sense substitutions of this sort include the

[9] E.g., Lakoff (1987). [10] Jackendoff (1997). [11] Nunberg (1979, 1995).

capital for the country's government as in (25), a place name for an event as in (26), and a place name for a typical product (27).

(25) But it was unclear whether Beijing would meet past UN demands for unrestricted access to [. . .].

(cited from Markert and Nissim 2006: 159, British National Corpus)

(26) At the time of Vietnam, increased spending led to inflation and a trade deficit.

(*ibid.* p. 158)

(27) We drank a nice Bordeaux last night.

The associations underlying such metonymies are lexicalized, and they may feature in the lexicon as co-existing senses of a word meaning—Bordeaux: 1) region of France; 2) type of wine. These are cases of "regular metonymy" because they reflect recurrent, entrenched conceptual mappings such as part for whole, cause for effect, person for role, place for event, etc.

However, metonymies such as the ones illustrated in (22)–(24) cannot be considered lexicalized, although they use the same mental processes. Many of them are produced and comprehended online; that is, they are novel and their interpretation depends on the concrete context of the utterance together with linguistic and pragmatic principles of interpretation. These utterances can be deemed unacceptable if the context in which they are produced is not taken into account, such as in the case of (22). We could consider them "novel" or "productive" metonymy, being mindful of the fact that regular and novel metonymy are not mutually exclusive, but rather two opposites on a cline of metonymy lexicalization.[12]

[12] Comprehension of metaphor is an even more productive process. The boundaries between metaphor and metonymy may be blurred; they may also be combined, as the following example suggests. "Sacred cow" is a metaphor, while "the streets of Stockholm" is a metonymy.

(i) The streets of Stockholm are awash with the blood of sacred cows. (The Economist)
This sentence refers to the time when the Swedish government decided to privatize and outsource education, health care, and care for the elderly. Note how much cultural and circumstantial knowledge is necessary to interpret the sentence correctly.

What happens when acquiring this lexical process in a second language? Different linguistic approaches agree that metonymy calculation is a universally available mechanism, although it is dependent on conceptual structures, cultural knowledge, and pragmatic routines. We are in familiar territory again: the mechanism is universal, so comprehending metonymic transfer of meaning should be easy for second language learners (once they know the words involved). However, comprehension involves computation over and above mere lexical access, including consideration of the context (sometimes cultural context), so one might expect to see some higher processing costs incurred. Existing research (Rundblad and Annaz 2010) suggests that metonymy comprehension in L1 English develops over time and reaches ceiling around the age of 12. This age of acquisition is considerably later than the ages when functional morphology or syntax are acquired. In an eyetracking study, Frisson and Pickering (2007) looked at the effect of familiarity on the processing of the producer for product regular metonymic pattern. They found that familiar metonyms (e.g., 'read Dickens') were straightforward to process, but unfamiliar metonyms (e.g., 'read Needham') caused processing difficulty unless context made it clear that the metonymic interpretation would be appropriate.

If children do not comprehend metonymy at ceiling until 12, it is a safe bet that L2 learners will also have a steep hill to climb. Among the notable recent studies are Chen and Lai (2011), which looks at Chinese native speakers' awareness of figurative language in L2 English, and Littlemore, Trautman Chen, Koester, and Barnden (2011), looking at nonnative undergraduates at British universities and their comprehension of academic texts. The latter authors analyzed the sources of difficulty and found that, among the expressions composed of words known to the participants, around 40% were metaphoric expressions. Slabakova, Cabrelli Amaro, and Kang (2014) investigated knowledge of regular and novel metonymy by Spanish and Korean learners of English. They discovered that the learners were influenced by their native patterns but the advanced participants were extremely proficient with metonymy calculation, having overcome native biases. Research on metonymy and more generally figurative language comprehension and use in the second language is still scarce, but very needed.

Teaching relevance

Research on figurative language computation suggests that non-compositional meaning enrichment increases the cognitive load of L2 speakers and leads to errors of understanding. It is very possible that these breakdowns of communication remain undetected in language classrooms and even university lecture halls. Furthermore, although the mechanism is universal, different languages use metonymy to different extents and with different frequency. Thirdly, the novel metonymy cases in the Slabakova at al. study were not highly rated by the native speakers, implying that such examples may be rare in the input to learners. All of these factors make it indispensable for teachers to understand the process of transfer of reference/metonymy/figurative language and to be able to expose learners to it.

9.6 Conclusion

The acquisition of the lexicon is under-investigated, from a pedagogical perspective. While we know a lot about how lexical items are stored in the mental lexicon of bilinguals and how they are accessed (sections 9.1 and 9.2), we do not seem to know enough to make the acquisition of a second language lexicon an easier, or at least a more manageable task. Exercise 9.2 contains one example of unnatural sounding learner speech but other examples abound. The research discussed in Section 9.3 goes some way towards addressing the difficulty of the acquisition task, thus making the work of language teachers more effective. This type of research promotes the idea that lexical meanings are composed of smaller semantic primitives, and that there are regular mappings between some of these semantic primitives, theta roles, and syntactic categories (subject, object). If verbs are learned in classes, instead of as chaotic assemblies of unstructured lists, acquisition will be more manageable and more successful. Similarly, exposing learners to regular as well as novel, productive metonymy and figurative language will enrich their linguistic repertoire profitably.

9.7 Exercises

Exercise 9.1. Take a lexical decision task in English. The link below provides three tasks, and you can try them all.

http://www.intro2psycholing.net/experiments/visual/vis_expt_index.php
Here is another type of lexical decision task:
http://psytoolkit.gla.ac.uk/library/ldt/

Exercise 9.2. *Everything is Illuminated* by Jonathan Safran Foer (2002).

Jonathan Safran Foer's debut novel tells the story of a young writer named Jonathan Safran Foer, who travels to Ukraine. The narrative alternates between his voice and that of his Ukrainian translator Alexander Perchov, a young man with a creative use of the English language. In this opening section, Alex introduces himself and prepares to meet his American visitor.

My legal name is Alexander Perchov. But all of my friends dub me Alex, because that is a more flaccid-to-utter version of my legal name. Mother dubs me Alexi-stop-spleening-me!, because I am always spleening her. If you want to know why I am always spleening her, it is because I am always elsewhere with friends, and disseminating so much currency, and performing so many things that can spleen a mother. Father used to dub me Shapka, for the fur hat I would don even in the summer month. He ceased dubbing me that because I ordered him to cease dubbing me that. It sounded boyish to me, and I have always thought of myself as very potent and generative. I have many many girls, believe me, and they all have a different name for me. One dubs me Baby, not because I am a baby, but because she attends to me. Another dubs me All Night. Do you want to know why? I have a girl who dubs me Currency, because I disseminate so much currency around her. She licks my chops for it. I have a miniature brother who dubs me Alli. I do not dig this name very much, but I dig him very much, so OK, I permit him to dub me Alli. As for his name, it is Little Igor, but Father dubs him Clumsy One, because he is always promenading into things. It was only four days previous that he made his eye blue from a mismanagement with a brick wall.

. . .

When we found each other, I was very flabbergasted by his appearance. This is an American? I thought. And also, This is a Jew? He was severely short. He wore spectacles and had diminutive hairs which were not split anywhere, but rested on his head like a Shapka. (If I were like Father, I might even have dubbed him Shapka.) He did not appear like either the Americans I had witnessed in magazines, with yellow hairs and muscles, or the Jews from history books, with no hairs and prominent bones. He was wearing nor blue jeans nor the uniform. In truth, he did not look like anything special at all. I was underwhelmed to the maximum.

Extracted from Jonathan Safran Foer's "Everything is Illuminated," published in the US by Penguin and in the UK by Harper Perennial.

Question 1: Why does this monologue sound unnatural in English? Is it because the construction of sentences is unacceptable, or is it that the lexical items are used in unacceptable combinations? Do you have trouble understanding what the intended message is?

See an excerpt of the movie (directed by Liev Schreiber) where Jonathan and Alex meet: https://www.youtube.com/watch?v=id15tjDK3K0.

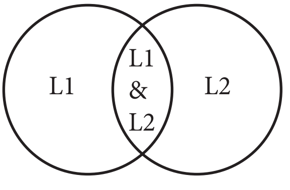
Question 2: Go through the excerpt and make lists of unacceptable choices of words and grammatical errors. Which list is longer?

Exercise 9.3. Re-read the discussion of unaccusative and unergative verbs in the chapter. Then consider the following adjectival uses of perfect participles:

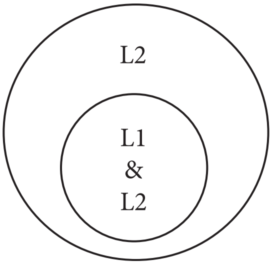
(i) fallen leaves, sunken ships, wilted lettuce, increased prices, escaped convicts
(ii) *worked people, *sung children, *dined people, *thought philosophers
(iii) murdered people, stolen books, destroyed buildings, rebuilt houses

Question 1: Explain informally what the above data may tell us about adjectival participles. How do the verbs in (i), (ii), and (iii) differ? What theta role (Agent or Theme) do adjectival participles modify? Do these examples support what we discussed in the chapter?
Question 2: Make a prediction for the L2 acquisition of these particles.

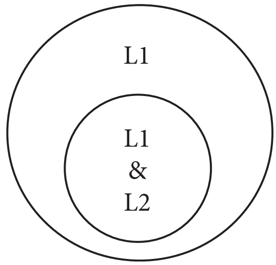
Exercise 9.4. Consider the three possible learning situations (illustrated below) in which the L2 input partially fits the L1 grammar.



Case A: *Partial overlap between L1 and L2*



Case B: *Subset L1–superset L2*



Case C: *Superset L1–subset L2*

Give examples of all three learning situations, numbering and explaining them clearly. (You can use studies discussed in previous chapters and exercises.) You may decide to give examples from argument structure, from syntactic properties, or from morphological properties. A learning principle has been proposed to describe Case B and Case C, it is called the Subset Principle. It states that learning in Case B (expanding the grammar) is easier than in Case C (shrinking the grammar). Use the unavailability of negative evidence in L2 acquisition to explain the Subset Principle. Why doesn't it apply to Case A? In which case is initial L1 transfer expected and why? What would you predict will happen in the later interlanguage development for each situation? Be sure to discuss negative and positive evidence, undergeneralization and overgeneralization.

Exercise 9.5 (do after Exercise 9.4). This exercise is based on the Inagaki (2001) study mentioned in the chapter. Try not to read the study before you tackle all the questions below. We will consider motion verbs with locational/directional PPs in English and Japanese. *Walk* and *run* are considered examples of manner-of-motion verbs while *go* and *come* are directed motion verbs.

(1) a. *John walked to school.*
b. *John ran into the house.*
c. *John went to school walking.*
d. *John went/came into the house running.*

(2) a. ?**John-ga gakkoo-ni aruita.*
John-NOM school-at walked
'John walked to school.'

b. ?**John-ga* *ie-no* *naka-ni* *hasitta.*
John-NOM house-GEN inside-at ran
'John ran into the house.'

c. *John-ga* *arui-te* *gakkoo-ni* *itta.*
John-NOM walk-GER school-at went
'John went to school walking.'

d. *John-ga* *hasit-te* *ie-no* *naka-ni* *itta/haitta.*
John-NOM run-GER house-GEN inside-at went/entered
'John went into/entered the house running.'

(NOTE: The Japanese examples contain one PP, where both the NP (subject) and the PP are case-marked.)

Question 1: In which of the three cases illustrated in Exercise 9.3 above (A, B, or C) does this argument structure property fall? Which language allows a broader range of motion verbs to appear with a goal PP? Draw a figure similar to the three above to illustrate the learning situation.

There is another contrast between English and Japanese in this domain: In English, manner-of-motion verbs with PPs involving such Ps as *under*, *behind*, and *in* allow either a locational or a directional reading (3), whereas their Japanese counterparts allow only a locational reading (4).

(3) a. *John swam under the bridge.* (directional/locational)
b. *John ran behind the wall.* (directional/locational)
c. *John jumped in the water.* (directional/locational)

(4) a. *John-ga* *hasi-no* *sita-de* *oyoida.* (locational only)
John-NOM bridge-GEN under-at swam
'John swam under the bridge.'

b. *John-ga* *kabe-no* *usiro-de* *hasitta.* (locational only)
John-NOM wall-GEN back-at ran
'John ran behind the wall.'

c. *John-ga* *puuru-no* *naka-de* *tonda.* (locational only)
John-NOM pool-GEN inside-at jumped
'John jumped in the pool.'

Question 2: Is there a subset–superset relationship here? Explain and illustrate.

Hypotheses for the study

Question 3: Formulate working hypotheses about the L2 acquisition of these properties by Japanese speakers learning English and English speakers learning Japanese. Explain where your hypotheses stem from. You are asked to use the relevant generative SLA theories and learning principles to formulate the hypotheses.

Participants

A bidirectional study was designed involving two sub-studies, one on Japanese speakers' acquisition of English as a second language (the ESL study) and the other on English speakers' acquisition of Japanese as a second language (the JSL study). Since each participant completed both English and Japanese versions of the experimental tasks, the Japanese participants in the ESL study served as a control group in the JSL study and the English participants in the JSL study served as a control group in the ESL study.

The ESL study compared a group of Japanese-speaking learners of English to a group of English controls. The learner group consisted of 47 first-year university students at Osaka Prefecture University who were majoring in engineering. They were divided into different proficiency levels (low and high intermediate), based on their scores on two sections (grammar and vocabulary) of the Michigan test. The control group consisted of 48 native speakers of English who were undergraduate or graduate students at the University of Hawaii.

The JSL study compared a group of English-speaking learners of Japanese to a group of Japanese controls. The learner group was the same as the control group of 48 English speakers in the ESL study. They were divided into three different proficiency levels (low intermediate, high intermediate, advanced) based on their scores on a proficiency test that was adapted from the Japanese Language Proficiency Test, Level 3. The control group consisted of 47 native speakers of Japanese, who constituted the learner group in the ESL study.

Test Instruments

A written picture-matching task was used for the ESL study. (See Figure 9.1, Inagaki's Appendix A for an example test item. The appendices can be found at the very end of this exercise.) Each test item consisted of an English sentence containing a manner-of-motion verb with a PP that was ambiguous between locational and directional readings. A pair of pictures, one of which showed a directional context and the other a locational context, followed the test sentence. In each picture, there were two objects—an object that moves, or a "figure," and an object with respect to which the figure moves, or the "ground". For example, in Appendix A (Figure 9.1), *Tom* was the figure and *bridge* was the ground. Both the figure and the ground were named in English to make sure that participants were familiar with the vocabulary. Participants were told that all pictures showed situations that took place in the past, and thus that all sentences would be in the past tense. Below each sentence were three options, *1 only*, *2 only*, and *either 1 or 2*. Participants were asked to circle *1 only* if the sentence matched the first picture only, *2 only* if it matched the second picture only, and *either 1 or 2* if it matched either the first or the second picture.

There were eight target items consisting of five manner-of-motion verbs (*walk, run, swim, crawl, jump)* and four prepositions *(in, on, under, behind)*. There were also seven distractors including both ambiguous and unambiguous sentences. To control for possible ordering effects, the test items and distractors were presented in two random orders, with about half of the participants taking one version and the rest the other version. The two pictures within each item were also randomly ordered for the same purpose.

A similar type of picture-matching task was developed for the JSL study. (See Appendix C, Figure 9.3, for an example test item.) Japanese was written in standard Japanese script, a mixture of *kanji* (characters of Chinese origin) and *kana* (the Japanese syllabary). *Kanji* characters were accompanied by *furigana* (a transliteration of *kanji* into *kana*) in order to ensure that participants had no difficulties comprehending the orthographic form of the sentences. Each test item consisted of a Japanese sentence containing a manner-of-motion verb with a PP headed by *de* 'at,' which is unambiguously locational, unlike its English counterpart, which is either locational or directional. In Figure 9.2, Inagaki's Appendix C, the test sentence *Tom-wa hasi-no* *sita-de aruita* 'Tom walked under the bridge' is followed by directional and locational pictures. Again, the test items and distractors were presented in two random orders, with about half of the participants taking one version and half the other. The two pictures within each item were also randomly ordered.

Question 4: Comment on the study design. In particular, do you find the task appropriate and why? What about the use of distractors and the two orders of the test? Have a look at the test sentences and comment on their number and naturalness. Is there anything in these test instruments that you would do differently? Be critical, and remember that all studies can be improved: no study is perfect.

Results

The ESL Study

Table 9.1 Mean responses by Japanese and English speakers in percentages (Standard Deviations in parentheses)

	Loc. only	**Dir. only**	**Loc./Dir.**
Low intermediate	69.92 (21.73)	7.42 (12.24)	22.66 (21.17)
High intermediate	76.67 (19.97)	5.83 (9.29)	17.50 (18.18)
English	25.52 (23.34)	8.33 (16.58)	66.15 (26.16)

Question 5: Illustrate the same results with a figure, using Excel or any other graph-creating software. Do not include the standard deviations in the figure.

A one-way ANOVA (analysis of variance) revealed that within responses of *locational only*, there was a significant effect of proficiency levels, $F(2, 92)=52.11$, $p=.0001$. Scheffé tests revealed that both the low- and high-intermediate Japanese groups differed significantly from the English group ($p = .0001$), but that they did not differ from each other. Furthermore, another one-way ANOVA showed that within responses of *either locational or directional*, there was a significant effect of proficiency levels, $F(2, 92)=44.15$, $p=.0001$, with Scheffé tests revealing significant differences between the English group and both of the two Japanese groups, which did not differ from each other.

Question 6: What do these results indicate with respect to your hypotheses?

Table 9.2 Number of Japanese and English participants answering either directional only or either locational or directional

Frequency of either *Dir. only* or *Loc./Dir.* responses (*k*=8)	**Low intermediate (*n*=32)**	**High intermediate (*n*=15)**	**English (*n*=48)**
0 – 3	24	14	5
4 – 6	8	1	22
7 or 8	0	0	21

We now turn to individual results to see if the group results above indeed reflect how participants of each group performed individually. Table 9.2 presents the number of participants in each group who answered *directional only* or *either locational or directional* a certain number of times. Responses of these two options are combined because what is crucial here is whether or not the participants recognized the directional reading of the target sentences.

Question 7: Describe what we see in Table 9.3. Do these individual results support the group results and why, or why not? Is your hypothesis for the ESL study confirmed or refuted?

The JSL Study

Table 9.3 Mean responses by English and Japanese speakers in percentages (Standard Deviations in parentheses)

	Loc. only	**Dir. only**	**Loc./Dir.**
Low intermediate	54.12 (35.19)	23.53 (24.73)	22.35 (22.23)
High intermediate	65.26 (32.55)	15.79 (17.10)	18.95 (28.65)
Advanced	68.33 (39.51)	13.33 (19.70)	18.33 (30.10)
Japanese	92.77 (12.11)	2.55 (7.93)	4.68 (9.52)

Question 8: Illustrate the same results with a figure. Do not include the standard deviations in the figure.

A one-way ANOVA shows that within responses of *locational only*, there was a significant effect of proficiency levels, $F(3, 91)=11.64$, $p=.0001$. Scheffé tests revealed that the control group was significantly different from all the learner groups ($p < .05$), which did not differ from each other. Within responses of *either locational or directional*, there was a significant effect of proficiency levels, $F(3, 91)=4.71$, $p=.0042$, with Scheffé tests revealing that the only significant difference existed between the low intermediate and the control groups ($p < .05$). Within responses of *directional only*, there was a significant effect of proficiency levels, $F(3, 91)=8.85$, $p=.0001$, with Scheffé tests revealing that the low- and high-intermediate groups were significantly different the control group, which did not differ from the advanced group.

Question 9: What do these results indicate with respect to your hypothesis?

Table 9.4 Number of English and Japanese participants answering either directional only or either locational or directional

Frequency of either *Dir. only* or *Loc./Dir.* responses (k=5)	Low intermediate (n=17)	High intermediate (n=19)	Advanced (n=12)	Japanese (n=47)
0	3	5	6	33
1 or 2	8	8	2	14
3 or 4	3	5	2	0
5	3	1	2	0

Table 9.4 presents the number of participants in each group who answered either *directional only* or *either locational or directional* a certain number of times. Responses of these two options are combined.

Question 10: Describe what we see in Table 9.4. Do these individual results support the group results? Why or why not? Is your hypothesis for the JSL study confirmed or refuted? Explain how, and refer to the data in the tables and figures.

Discussion

Question 11: Discuss the results of the two studies together. What do these results tell us about the learning principle called the Subset Principle and the nature of the input in second language acquisition? Is one of the two learning directions investigated in this study more successful than the other, and why might that be? Conclude by stating the central finding of this study.

Question 12: Discuss what the results of this study tell us about L1 transfer in L2 acquisition. If you think that there is evidence of L1 transfer in the data, discuss how it affects the pace of L2 acquisition (refer to the different proficiency groups).

Question 13: What are the implications of this study for the contemporary theories of adult second language acquisition?

Inagaki's (2006) picture-matching tasks

Appendix A

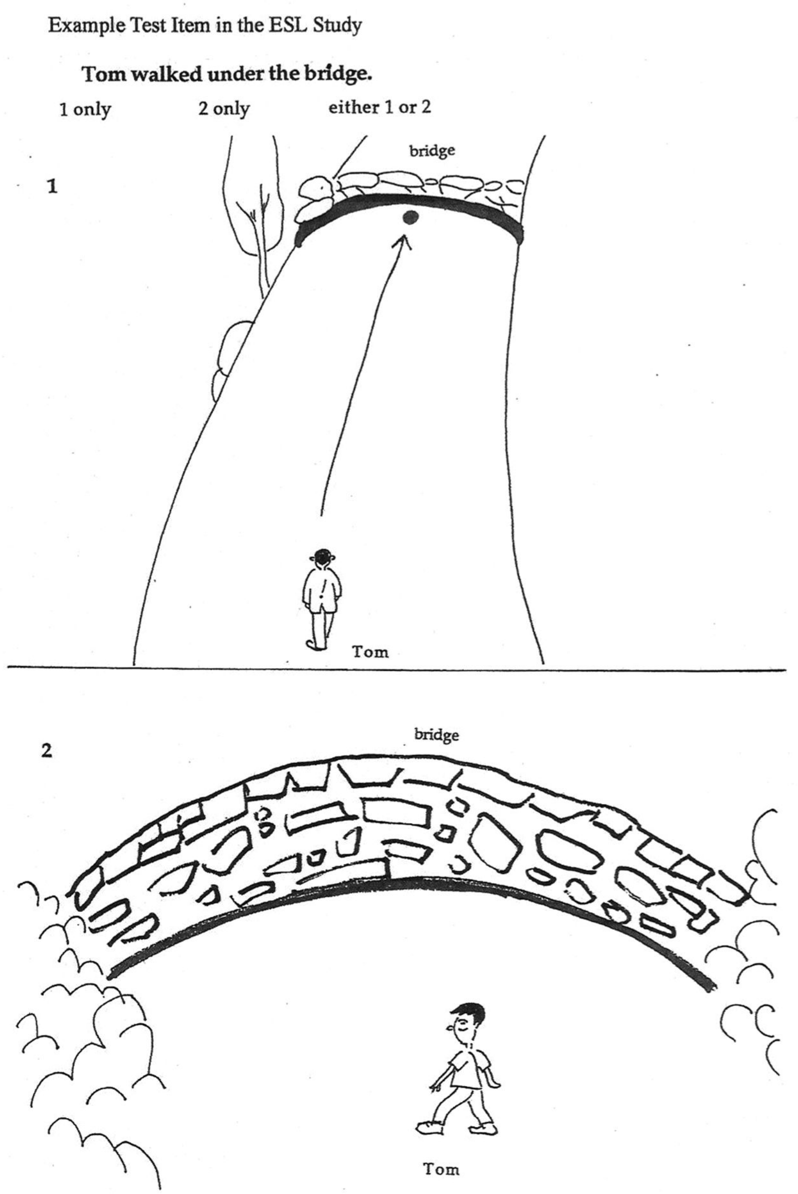


Figure 9.1 Appendix A, from Inagaki (2006)

Appendix B

Sentences Included in the ESL Picture-Matching Task

A: Test sentences

1. Jim walked behind the house.
2. Tom walked under the bridge.
3. Ted ran behind the wall.
4. Mary ran on the stage.
5. Peter swam in the cave.
6. The baby crawled under the table.
7. The mouse crawled on the table.
8. Fred jumped in the pool.

B: Distractors

Directional only

1. Sam walked to the beach.
2. John ran into the gym.
3. Paul jumped onto the bed.

Locational only

4. Jim was in the park.
5. John ran at the racetrack.

Ambiguous

6. Tom watched the man with binoculars.
7. John saw fat cats and dogs.

Appendix C

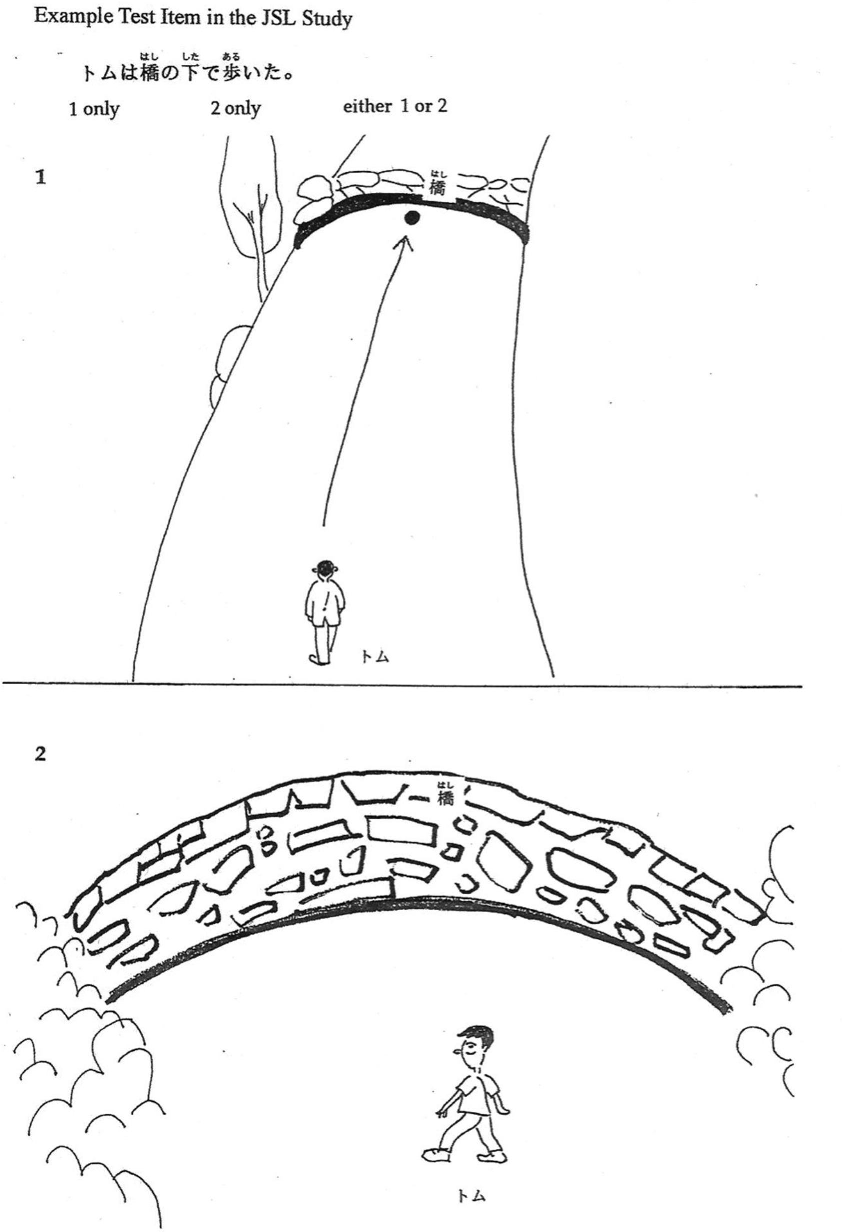


Figure 9.2 Appendix C, from Inagaki (2006)

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Appendix D

Sentences Included in the JSL Picture-Matching Task

A: Test sentences

1. *Tom-wa hasi-no sita-de aruita.*
   Tom-TOP bridge-GEN under-at walked
   'Tom walked under the bridge.'
2. *John-wa taiikukan-no naka-de hasitta.*
   John-TOP gym-GEN in-at walked
   'John ran in the gym.'
3. *Akachan-wa ie-no ura-de hatta.*
   baby-TOP house-GEN behind-at crawled
   'The baby crawled behind the house.'
4. *Paul-wa beddo-no ue-de tonda.*
   Paul-TOP bed-GEN on-at jumped
   'Paul jumped on the bed.'
5. *Hikoosen-wa sima-no ue-de tonda.*
   blimp-TOP island-GEN over-at flew
   'The blimp flew over the island.'

B: Distractors

Directional only

1. *Sam-wa hamabe-ni aruite itta.*
   Sam-TOP beach-at walking went
   'Sam went to the beach walking.'
2. *Mary-wa steezi-no ue-ni hasitte agatta.*
   Mary-TOP stage-GEN on-at running went-up
   'Mary went onto the stage running.'
3. *Peter-wa dookutu-no naka-ni oyoide haitta.*
   Peter-TOP cave-GEN in-*ni* swimming entered
   'Peter entered the cave swimming.'

Ambiguous

4. *Mary-wa* *Paul to* *Tom-no* *otoosan-ni* *atta.*
   Mary-TOP Paul and Tom-GEN father-DAT met
   'Mary met Paul and Mary's father.'

5. *John-wa* *Mary-ga* *sukidatta.*
   John-TOP Mary-NOM loved
   'John loved Mary' or 'John, Mary loved.'